

# Solvency Field Theory V13.13

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What Exists, What Is Relational, and What Is Bookkeeping

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A Provenance-Governed Ontology Constitution for the V13 Stack

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**V13.13 is a ground state, not a ceiling.**

Status: FIRST PUBLIC DRAFT v0.1 - NOT FROZEN - ASSEMBLED FROM THE AUDITED V13.13 ONTOLOGY PACK

*August 2026*

## Abstract

Solvency Field Theory developed its particle, gravity, expansion, quantum, clock, ruler, electron, and interaction programs in separate documents. Across V13.0 through V13.12 those programs gradually converged on a smaller and more consequential claim about reality itself. The present paper assembles that claim into one provenance-governed ontology.

The current SFT ground state contains real causal propagation, persistent recurrent physical structures, local joint interactions, and irreversible realized four-dimensional history. It does not contain a physical cosmic ledger, a memory surface, a universal present, a microscopic write-address substrate, or an ontic write operator. A maintained particle is the recurrent structure itself rather than a carrier inside a container. One persistent object may possess invariant magnitudes; separation, relative components, interactions, force, entanglement, and measurement belong to physical relations. A lawful interaction is therefore a complete constrained joint configuration, with force appearing as a participant-side readout relative to isolated continuation rather than as a third substance. Causal structure supplies shape without an intrinsic ruler; a persistent closure supplies a local rest frame, tally, and unit; two persistent things make comparison possible; many persistent things make extension and macroscopic apparatus possible.

This paper does not claim that every sentence has the same proof status. It separates controlling ontology, model freezes, exact conditional results, scoped proposals, candidate extensions, open joints, and retired interpretations. It preserves V13.10's completed-history account of time as a definitional proposal rather than silently promoting it to theorem. It preserves the post-interaction persistence criterion, structural identity, open-amount/closed-instance asymmetry, magnitude-at-one/calibration-at-two synthesis, and relational reading of curvature as visible candidate programs rather than constitutional facts. It also names the unresolved actualization, composition, formation, null-transfer, coarse-graining, expansion, scale, and coupling problems.

V13.13 is deliberately amendable. It controls interpretation by default, but it has no veto authority over later discovery. A compatible extension expands the ontology; a refinement retypes it; a contradiction opens an ontology challenge audit. If a later result survives hostile review, this document must change. The strongest methodological sentence is therefore also the final one: **SFT is constrained by reality, not by SFT V13.13.**

**Strongest honest claim. V13.13 does not derive a new universe from prose. It records, with explicit provenance and boundaries, the smallest coherent ontology currently justified by the V13 stack.**

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## Document status and inheritance rule

This paper is an integration document. It does not replace the technical derivations owned by the particle, gravity, expansion, entanglement, clock, ruler, electron, or interaction sectors. It controls ontology only where it explicitly adjudicates a claim and identifies the source that owns it. The unit of inheritance is the **claim or section**, not the whole paper. One source may contain portable mathematics, obsolete ontology, useful negative results, and unresolved debt at the same time. [S09, S15]

The controlling center of the present ontology is V13.8-V2 for the write/operator correction, V13.9-V3 for Level-2 relational structure, V13.11 for the level and ruler ladder, V13.12 for the maintained-electron ontology, and V13.4-V2 for complete joint interactions. V13.10 contributes a bounded definition/proposal concerning time. V13.0 through V13.2 remain indispensable ancestry, but their literal write, ledger, and writable-address language must be translated or rederived under the current ontology. No separate controlling ontology source was identified in the audited library under the V13.3 or V13.6 labels; this paper does not invent missing doctrine merely to make the numbering continuous. [S01-S15]

**No narrative laundering.** A statement does not become constitutional because it is memorable, elegant, or repeated across sectors. It enters the core only at the status and scope earned by its source.

### Status vocabulary

| Status            | Meaning in V13.13                                                                  |
|-------------------|------------------------------------------------------------------------------------|
| CONTROLLING       | Current default ontology or methodology at the stated scope.                       |
| MODEL FREEZE      | Frozen within a declared model or representation; not automatically universal.     |
| EXACT CONDITIONAL | Exact inside explicit premises and a bounded theorem domain.                       |
| SCOPED / PROPOSAL | Useful and auditable, but not a general theorem.                                   |
| CANDIDATE         | Promising synthesis or theorem target that remains challengeable before promotion. |
| OPEN              | Not earned and prohibited from promotion by rhetoric or resemblance.               |
| RETIRED / RETYPED | Old ontology removed; surviving content must cite a later independent owner.       |
| METHODOLOGICAL    | Rule governing provenance, testing, interpretation, or supersession.               |

## The constitutional sentences

The following eleven sentences are the compact public face of the present ontology. Their status is not uniform. Some are controlling across the stack, some contain an explicit model-scope clause, and the final sentence governs future amendment. The detailed chapters and provenance appendices state those boundaries.

**Real physical support propagates causally; at the primitive local level SFT models that propagation as null.**

**Persistent massive things are recurrent closures of real physical support, and a maintained particle is the recurrent structure itself.**

**Within the maintained-electron model, finite support, source, geometry, spin/framing, charged dressing, recurrence, mass, and clock are faces or readouts of one maintained state rather than separate objects bolted together.**

**Recurrence supplies persistent identity, a local rest frame, tally, and unit; the clock is a post-closure readout, not a prerequisite for closure.**

**Realized four-dimensional history is irreversible, but there is no physical substrate, ledger, memory surface, or operator on which reality is stored or appended.**

**One thing can possess intrinsic magnitudes; separation, relative components, interactions, and measurements belong to physical relations.**

**Relations can contain genuine physical structure, and a lawful joint configuration need not factor into independently complete local parts.**

**Interactions are complete constrained joint configurations; force is a participant-side readout of the difference between joint and isolated continuation.**

**Causal geometry supplies shape without size; persistent closure supplies a local unit; two persistent things supply comparison; many supply extension.**

**Projectors, transition maps, fields, metrics, and response operators must be interpreted according to independently earned physical content, not reified merely because the mathematics is useful.**

**V13.13 records SFT's current ontological ground state, not a ceiling: a later result that survives hostile review may amend or supersede any claim herein.**

**One thing can have magnitudes. Two things have relationships. Measurement asks what that relationship makes observable.**

## 1. The question V13.13 answers

The question is not whether SFT has accumulated enough models to deserve a summary. It is whether those models now imply a coherent answer to a more fundamental question:

**What does SFT currently say is physically real, what exists only in relation, and what belongs to our mathematical bookkeeping rather than to nature?**

V13.0 began with a one-support thesis: light and mass were not treated as two substances, but as open and closed organizations of real causal support. V13.1 and V13.2 attempted to extend that same foundation into gravity and expansion. V13.5 through V13.9 pressed the quantum and entanglement sectors. V13.10 asked what a clock is doing; V13.11 asked where a ruler comes from; V13.12 built a maintained electron whose claimed readouts coexist on one state; V13.4-V2 then rebuilt force and reaction around complete joint configurations. Finally, V13.8-V2 removed the literal write machine that had once supplied a common metaphor for all of them. [S01-S14]

That correction did not leave a hole. It exposed the physical order that had been present underneath the representation:

physical configuration → lawful solution → mathematical restriction or response → readout .

Earlier SFT often reasoned in the opposite direction. A projector, write count, lane split, or transition map was introduced as a useful representation and then allowed to carry microscopic explanatory weight. The newer stack reverses that move. The physical configuration comes first. The mathematics earns its interpretation afterward. [S09]

The result is smaller because it removes hidden machinery. It is larger because the same ontology now organizes particles, clocks, rulers, interactions, entanglement, gravity, and measurement.

## 2. What is physically primitive

### 2.1 Real propagation rather than a background medium

The earliest current SFT layer is not a material spacetime medium, an aether, a lattice, a foam, or a ledger. It is real causal propagation. At the present SFT particle scope, the primitive local propagation is modeled as null. This does not mean every physical system is a free photon. It means the local causal support does not begin from microscopic rest. Open support propagates outward; persistent massive support organizes locally null propagation into a recurrent global structure. [S01, S09, S10, S13]

This is the surviving physical content of the older phrase “one kind of stuff.” The modern form is deliberately less metaphorical. It does not require an energy ocean, storage medium, or invisible machinery behind the support. It states only that light and maintained matter are not introduced as ontologically unrelated substances. They are different lawful organizations of real causal support. [S01]

### 2.2 Irreversible history without memory hardware

The second primitive is realized history. What physically occurred cannot later become a different event. That is the current meaning of no-overwrite. It is a statement about actuality and causal history, not about occupied cells in a universal database. [S09, S10]

A history tally can remain useful. A transition map can remain useful. The word “write” can remain useful when explicitly translated. None of those conveniences implies that the universe has somewhere to write. There is no cosmic filing cabinet that makes history irreversible by recording it. History is irreversible because it happened.

**Current no-overwrite: realized four-dimensional history is irreversible; what occurred cannot later become a different event.**

This distinction matters because it removes an unnecessary substrate while preserving the causal arrow. It also changes how expansion, time, measurement, and branch realization must be described. Any future mechanism that requires a finite store of unoccupied addresses, a material memory surface, or an operator that appends reality must re-earn that structure through independent physical evidence. It may not inherit it from old vocabulary. [S03, S09]

## 3. From propagation to a persistent thing

### 3.1 Open propagation and recurrent closure

Open null propagation and persistent recurrent massive structure are different physical branches. The difference is topological and dynamical, not a change from one substance into another. An open mode advances without supplying its own massive rest frame or recurrence tally. A maintained massive root returns through a finite recurrent physical configuration and thereby becomes a persistent thing. [S01, S12-S14]

The exact first-capture law remains open. This paper does not use the existence of a maintained particle to explain how that particle formed. The mature clock, ruler, support normalization, and invariant root identity are downstream outputs of closure. They may not be fed backward into the formation dynamics as selectors. [S13, S14]

### 3.2 The particle is the structure

The maintained-electron program removed several pictures that once looked like physical mechanisms: a photon moving inside a container, literal six-slot storage, an independent hard wall, scalar shape used as spin, and a separate clock mechanism. The remaining object is one finite recurrent state. Its local null flow, transverse support, source/current, self-consistent metric response, spin/framing, charged dressing, recurrence, and clock readout are different faces of that one maintained structure. [S13, S14]

This claim is controlling at Electron V1 scope. It is not automatically a theorem that every possible particle species must possess the identical constitutive architecture. The broader ontology that survives is more modest and more useful:

**A maintained particle is not required to contain a smaller object that performs the particle's behavior. The maintained physical configuration may itself be the particle.**

### 3.3 Rest without microscopic stillness

The electron model supplies a concrete example of rest without local stasis. Local support remains null while the complete recurrent route closes and its integrated translational momentum vanishes. A persistent rest frame can therefore be a global property of recurrence rather than a statement that the microscopic primitive has stopped. [S13, S14]

That distinction is important for the one-speed ancestry. It allows SFT to treat massive rest as an organized causal process instead of smuggling in a second kind of stationary substance. It also clarifies the difference between a locally null tangent and a globally timelike persistent object.

### 3.4 Mass and charge are not the same handle

Mass is interpreted as an invariant/readout or response of maintained closure, not as stored material attached to a carrier. Charge/framing is a separate root handle. The phase that marks rest-mass cadence and the phase that marks signed charge/framing cannot be collapsed into one variable: same-charge particles can have different masses, and equal-mass particle-antiparticle partners can have opposite charge. [S05, S09]

The distinction does not derive the numerical value of charge, charge quantization, alpha, or the full species map. It states that SFT must preserve at least two independent physical handles in any adequate particle representation.

### 3.5 Root identity and surplus content

A branch-preserving process does not continuously turn one electron root into “more electron.” Additional conserved energy can appear as relative four-momentum, fields, radiation, excitation, or access to additional channels while the root retains its invariant branch identity. This is the root-surplus distinction. [S05]

persistent root invariant content  $\neq$  surplus relational, kinematic, field, radiation, or channel content .

The distinction is earned at branch-preserving scope. It does not yet provide a native creation/annihilation theorem or a complete field-quantization ontology.

## 4. The level ladder

The V13.11 ladder orders the ontology by what must physically exist before a class of facts becomes meaningful. The ladder is not a claim that nature performs four numbered software steps. It is a dependency map.

| Level          | Physical threshold                      | What becomes meaningful                                                                                                | Still absent or restricted                                                                         |
|----------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Level 0</b> | Null causal structure                   | Light cones, causal order, conformal geometry, null propagation, helicity.                                             | No massive rest frame, recurrence tally, local ruler, or relational component.                     |
| <b>Level 1</b> | One persistent closure                  | Persistent identity, rest frame, local tally/unit, mass, charge/framing, spin magnitude and other one-root invariants. | No separation, relative velocity, relative orientation, analyzer component, or mutual interaction. |
| <b>Level 2</b> | Persistent systems participate together | Separation, relative motion and orientation, interaction, measurement, joint nonfactorization, entanglement.           | A relation is physical but is not automatically a material substance or signal.                    |
| <b>Level 3</b> | Many persistent systems                 | Rigid extension, rods, collective gravitation, thermodynamics, horizons, macroscopic apparatus.                        | Detailed collective laws remain sector-specific.                                                   |

#### 4.1 Do not assign a higher-level fact to a lower-level object

The central rule is:

**Do not assign to one object what physically exists only in a relation.**

A lone persistent object may possess spin magnitude. A spin direction relative to an analyzer does not exist until the analyzer relation exists. A root may possess mass/source structure. Separation, mutual acceleration, redshift between clocks, and deflection relative to another path are relational facts. A closure may possess a local cadence. Time dilation becomes an observation only when histories are compared. [S10, S12]

This is not anti-realism. Relations can be fully physical without belonging to either participant separately. Nor is it observer dependence in the psychological sense. A relation can exist and evolve with no human present. Measurement is a physical comparison performed by an apparatus, not a mental act that creates reality. [S10, S12]

#### 4.2 The apparatus is constituted, not presupposed

Coordinates, frames, rods, clocks, metric scale, and macroscopic apparatus are often treated as the stage on which matter acts. The level ladder reverses that explanatory order. Causal structure supplies shape without a scale. Persistent closure supplies a local rest frame and unit. Two persistent systems permit comparison. Many persistent systems permit rigid extension and an apparatus. The language in which physics writes its equations exists because persistent physical things make that language operational. [S12]

### 5. Time, clocks, rulers, and anchors

#### 5.1 No universal present

There is no master clock or universal present shared by all systems. Cross-clock statements require a physical comparison relation. This is a controlling relational consequence of the V13.9-V13.11 stack, independent of whether every stronger sentence in V13.10 is ultimately retained. [S10-S12]

A local recurrent object does not consult a global register to know its history. Its tally is a local fact about its own persistence. The absence of a master clock is therefore not a missing coordination service; it is the expected result when elapsed histories belong to local processes.



## 5.2 The clock is downstream of closure

The first lawful particle clock appears only after persistent recurrence exists. A forming branch cannot use its eventual Compton period, rest frame, or ruler to choose the dynamics that create that branch. The chronology is:

open causal support → capture and first lawful closure → persistent recurrence → clock and ruler readout .

This dependency firewall is controlling even if the detailed formation mechanism remains open. [S13, S14]

## 5.3 Time as completed history - proposal status

V13.10 proposes that elapsed time is the normalized count of completed local recurrent history rather than an external universal flow. In that proposal, a clock does not estimate a separate substance called time; the completed recurrence is the clock's own elapsed-time history. [S11]

The proposal explains why different clocks can accumulate different durations without dividing one universal resource, why null propagation lacks an intrinsic massive tally, and why a count of history is not naturally represented by an ordinary instantaneous observable operator. But V13.10 explicitly states that it defines time rather than deriving it. V13.13 therefore preserves the proposal at scoped status.

**Scoped V13.10 sentence:** A clock does not merely measure an external time; its completed recurrence makes its own elapsed-time history.

## 5.4 Causal shape and the origin of scale

Null causal/conformal structure determines causal shape but not an intrinsic metric scale. It provides no massive rest frame and no local ruler. A persistent closed process supplies a local Compton-associated unit and thereby makes a local scale physically available. [S12]

The ruler comes after the thing capable of being a ruler. This is more than a slogan. It prevents SFT from using a downstream physical standard to normalize the upstream dynamics that create the standard.

## 5.5 Magnitude at one, calibration at two

The current core distinguishes magnitude from component. One persistent root can possess an invariant magnitude; a direction or component relative to another object belongs to the relation. A stronger synthesis is plausible but not yet constitutional:

**Candidate C-04:** One persistent object may possess an intrinsic magnitude; a numerical laboratory value and unit assignment require comparison to a physical standard.

This candidate is intentionally weaker than “numbers begin at two.” A Level-1 closure can possess a tally, cadence, and dimensionless invariants. The open question is how an intrinsic magnitude becomes an operational number in a shared unit system without erasing one-object facts. The measurement firewall adds a second rule: a later empirical anchor may identify or calibrate a structure, but it does not retroactively become a first-principles prediction or a mechanism that built the structure. [S12, S16]

## 6. Relations are physical

### 6.1 Joint continuation need not factor

Let ( $\_A$ ) and ( $\_B$ ) denote the lawful isolated continuation spaces of two persistent systems. After a genuine interaction, the lawful joint continuation need not be the Cartesian product of the isolated continuations:

$$\Omega_{AB} \neq \Gamma_A \times \Gamma_B.$$

This is a physical claim about the complete configuration, not a claim that a material relation-fluid has been added between the participants. [S05, S10]

Nonfactorization alone is not quantum entanglement. Classical conservation constraints can also produce nonproduct legal outcome sets. Bell-relevant structure additionally requires phase-bearing relational information that cannot be reconstructed from two independently complete local marginals. [S10]

### 6.2 Separation does not necessarily produce independence

Clean local separation can remove direct mechanical overlap without erasing nonfactorizable relational structure. In the V13.9-V3 response representation, invertible local transport preserves rank. Independence therefore requires a physical process that changes the relation or its accessibility; mere distance is not enough. [S10]

This reverses a common intuition. Entanglement is not treated as a mysterious payload that each particle carries away. The joint structure belongs to the pair configuration. The dynamical question is how distinguishing interactions redistribute or reduce direct pair accessibility until an effectively independent description becomes lawful.

### 6.3 Entanglement is Level-2 structure

At the frozen V13.9-V3 scope, entanglement is modeled as phase-bearing Level-2 joint structure rather than as a hidden answer stored in either particle. Local source and analyzer interactions, causal propagation, no-signaling, and no preferred spacelike measurement order are preserved. Bell-local separability of the complete relational source object is not. [S10]

The exact binary redistribution theorem is narrower than a universal conservation law. In the normalized pure binary branch-preserving distinguishing domain,

$$C_0^2 = A_{AB}^2 + X_{AB:E}^2,$$

where direct pair accessibility is reduced as relational structure becomes encoded in the larger system-environment configuration. General mixed-state, branch-mixing, and multipartite conservation remain open. [S10]

### 6.4 A relation is not a signal or substance

The language of relation, capacity, response, and accessibility is representational unless a particular physical content has been independently identified. None of those words automatically denotes a fluid, signal, hidden field, conserved charge, or material transmitted between particles. [S10]

Real physical support still propagates causally. The relational claim is not an excuse to bypass causal dynamics. It says that the complete physical fact may belong to a joint configuration even when no independent local payload carried by either participant reconstructs it.

## 7. Interaction without a force substance

### 7.1 The complete joint configuration

The mature V13.4-V2 architecture begins from persistent participants and one lawful joint domain. Schematically,

$$S_J[X] = \int_M L_{native}(X, \nabla X, g, \dots) d^4x, X \in D_J,$$

with physical evolution determined by

$$\delta S_J[X] = 0, X \in D_J.$$

The constitutive action and lawful domain define the physics. KKT projectors, Schur complements, effective actions, retarded inverses, and response operators describe restrictions or tangents of that already-defined physical system. [S05, S09]

### 7.2 Force as participant-side readout

Ordinary language says that object A exerts a force on object B. That language is operationally effective, but it can encourage an unnecessary ontology: two independently complete objects plus a third entity called force. SFT instead defines force as the participant-side change in lawful continuation produced by the complete joint configuration relative to isolated continuation. [S05]

**Force is the participant-side dynamical readout of a physical interaction: the change in a system's lawful continuation produced by the complete joint configuration in which it participates, relative to its isolated continuation.**

This definition is broader than proper force. In gravitational free fall the proper force can vanish while the worldline differs from its isolated flat-spacetime continuation. SFT therefore treats force as one readout class within interaction geometry rather than the deepest category.

### 7.3 Force, scattering, reaction, and measurement

The same architecture distinguishes several process classes without turning them into separate substances:

| Process                  | Native SFT description                                                               |
|--------------------------|--------------------------------------------------------------------------------------|
| <b>Force</b>             | Continuous identity-preserving participant readout.                                  |
| <b>Scattering</b>        | Finite identity-preserving joint transition.                                         |
| <b>Reaction or decay</b> | Identity-changing joint transition.                                                  |
| <b>Measurement</b>       | System-apparatus joint interaction producing distinguishable descendants or records. |

This common architecture does not mean electromagnetism, gravity, strong interaction, weak transition, detector completion, and every other sector share one microscopic constitutive kernel. It means they are all required to specify a lawful complete configuration rather than treating “the force” as an unexplained add-on. [S05]

### 7.4 The electron interaction firewall

The frozen maintained electron supplies an isolated Level-1 root target. A real interaction must solve a joint state that approaches the isolated roots in the large-separation boundary. It must not apply an invisible external force to a permanently isolated electron while leaving the source, support, field, and geometry configuration unchanged. [S05, S13]

Conversely, an interaction program that must retune or redesign the frozen isolated electron merely to make the interaction work has not automatically extended V13.12. It has challenged or superseded the parent model and must say so explicitly.

## 8. Geometry, gravity, and the substrate question

### 8.1 Geometry is not a rubber medium

Current SFT does not deny geometry, metrics, fields, or curvature. It denies that their mathematical usefulness automatically proves the existence of a deformable material substrate. Causal structure is physical structure. Metric scale is physical structure. The statement “geometry is not a material medium” is therefore not the statement “geometry is unreal.” [S09, S12]

The level ladder gives a constructive account of the apparatus of geometry. Null causal structure supplies conformal shape. Persistent closure supplies a local unit. Two persistent systems make separation and clock comparison meaningful. Many persistent systems make rods, extended apparatus, thermodynamics, and horizons possible. The spacetime language used by physics is assembled from causal and persistent physical structure rather than presumed as a furnished material stage. [S12]

### 8.2 Intrinsic source structure and mutual gravitational readout

A persistent root can possess invariant mass/source structure and a local closure-gravity or metric response. The maintained-electron model explicitly uses the same normalized source for its local structure without introducing a second gravitational mass stock. [S13]

Mutual acceleration, relative redshift, deflection, clock comparison, and other observed gravitational effects require a relation or a larger configuration. V13.11 therefore places mutual gravitation at Level 2. [S12]

The resulting distinction is:

intrinsic source and local response structure  $\neq$  mutual gravitational observable .

### 8.3 What V13.13 does not yet say about curvature

V13.13 does not declare that a single source has no metric solution, that spacetime curvature is nonexistent, or that all V13.1 exterior and two-body results have already been rederived in relational language. Those would be new claims beyond the audited stack.

It does preserve a high-value reaudit target:

**Candidate C-05:** A root carries source and intrinsic response structure; what is observed as bending, deflection, redshift, or mutual gravitation belongs to the lawful relational configuration rather than to a deformable material substrate.

The promotion gate is demanding. The V13.1 exterior and two-body observables must be reformulated in the current joint-configuration ontology while preserving every earned weak-field, relativistic, and conservation limit. If an independently measurable geometric degree of freedom remains that cannot be captured by the source-and-relation interpretation, the candidate must be retyped rather than protected.

### 8.4 Expansion after the write-operator correction

V13.2's positive target survives at candidate level: expansion is not required to be a new force, medium, or dark substance, and it may be a global readout of physical history/source dynamics. Its old mechanism language does not survive automatically. Writable-address budgets, universal fresh-address

demand, a cosmic capacity ledger, and “the lane the write could not settle” are retired as literal ontology. [S03, S09]

A post-V13.8 expansion theory must state its dynamics through current causal, geometric, source, horizon, and coarse-graining objects. The current ontology register therefore marks expansion for retyping and rederivation rather than pretending the old language is still controlling.

## 9. Mathematics and ontology

### 9.1 Operators remain mathematics

The death of the ontic write operator is not a ban on operators. Projectors can exactly represent complementary subspaces. A transition map can exactly represent the relation between lawful input and output configurations. A Schur complement can exactly represent tangent response after on-shell elimination. A field or metric can be indispensable to the physical description. [S05, S09]

The prohibited inference is:

useful mathematical object  $\Rightarrow$  hidden material machine executed by nature .

Every mathematical object requires a physical interpretation appropriate to its independently derived role. A projector may represent admissibility or tangent/normal decomposition. It does not automatically divide one microscopic event into literal settlement lanes. A repeated rank ratio may be mathematically exact. It does not automatically identify a universal physical fraction. [S09]

### 9.2 Solution maps are downstream

For a complete configuration ( $X=(x,y)$ ), responding degrees of freedom may be solved on shell as ( $y=y_.$ ). The effective participant functional and response operators are then derived from the complete action. The transition map is the solution map induced by that physics. It is not a second ontic machine that performs the evolution after the action has already done so. [S05, S09]

The controlling hierarchy remains:

**physical configuration -> lawful joint solution -> mathematical restriction/response -> observable readout**

### 9.3 Native-first correspondence

Where QFT, the Standard Model, GR, or standard mathematical terminology materially helps orientation, V13.13 uses a native-first correspondence rule:

**In QFT/SM or GR this structure is referred to as X; in SFT the native physical interpretation is Y.**

Standard mathematical names may be used freely when the structures genuinely coincide. What may not be imported automatically is the external theory’s ontology. QFT and GR remain precision and correspondence layers unless SFT independently earns the same physical interpretation. [S05]

### 9.4 One physical source counted once

The anti-reification discipline also blocks double counting. A self-field, metric response, Hilbert source, Belinfante source, or interaction field cannot be appended as a second positive copy of the same physical sector merely because several mathematical descriptions are available. The physical configuration is primary; equivalent descriptions must be reconciled rather than summed as independent substances. [S13, S15]

## 10. Persistence, identity, and instancehood

The level ladder describes what exists once a persistent closure has been obtained. It does not by itself explain how a temporary interaction becomes a new enduring thing or why repeated instances of one root class possess identical intrinsic properties. Those are the strongest candidate extensions produced by the present audit.

### 10.1 Candidate post-interaction persistence criterion

**Candidate C-01:** A new persistent thing is formed when a temporary joint configuration captures into a lawful continuation that remains self-maintaining after the initiating encounter ends.

The candidate has the right shape to unify particle formation, binding, record formation, chemistry, and phase change: a temporary joint configuration produces a descendant continuation that remains lawful after the initiating relation is removed. But that cross-domain generality is not yet earned. The first-capture dynamics remain open even in the particle program. The candidate must not use the downstream clock or identity of the completed object to cause its own formation. [S05, S13, S14]

Promotion requires a domain-independent persistence criterion tested in particle formation and at least one genuinely nonparticle persistence class without changing the definition between sectors. If different sectors require incompatible notions of persistence, the candidate becomes a family resemblance rather than a constitutional law.

### 10.2 Candidate structural identity

**Candidate C-02:** Identical roots are one recurrent solution class instantiated many times; same-class instances share invariant root properties.

The maintained-electron ontology strongly suggests structural identity. If every lawful instance belongs to the same unique recurrent root class, then its invariant root properties are fixed by the class rather than adjusted separately for each electron. This is a plausible route to why electrons share the same intrinsic public identity. [S14]

It is not yet a theorem of quantum indistinguishability. It does not derive exchange antisymmetry, Pauli exclusion, self-composition nullity, or the fermionic sign. Those remain owned by the P6-L local relativistic composition/statistics program. A structural-identity theorem must first prove that the relevant invariant root class is unique or physically equivalent across instances and that no hidden inequivalent branches reproduce the same public label with different intrinsic invariants.

### 10.3 Candidate open amounts and closed instances

**Candidate C-03:** Open modes come in continuously variable amounts; persistent closed roots come in instances, while surplus energy occupies motion, fields, radiation, excitation, or new channels.

The earned root-nonscalability result supports the closed-root side at branch-preserving scope. The open-mode side is a natural synthesis because amplitude and occupancy are meaningful for open propagation. The full asymmetry remains candidate until a native creation and composition law reproduces it without importing QFT ontology as a premise. [S05, S14]

A lawful continuous degree of freedom in the root itself would kill or retype the claim. So would evidence that the supposed amplitude is not merely relational, excitation, or environment content but changes the invariant root class continuously.



## 11. What current SFT explicitly does not posit

The anti-ontology list is not a list of topics SFT has forgotten. It records interpretations that the current stack retires, supersedes, or forbids from automatic promotion.

| Retired or prohibited interpretation                                                                                           | Current ruling                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| A physical cosmic ledger, universal memory surface, write-address database, or ontic write operator.                           | History is irreversible actuality; write and ledger are bookkeeping shorthand only. |
| A universal present or master clock.                                                                                           | Histories and clocks are local; comparison is relational.                           |
| A mandatory aether, lattice, foam, or material spacetime medium.                                                               | Causal and metric geometry are structure, not automatically material substrate.     |
| A photon or carrier living inside a separate particle container.                                                               | At Electron V1 scope, the recurrent maintained configuration is the particle.       |
| Mass as stored substance attached to a carrier.                                                                                | Mass is an invariant/readout or response of maintained closure.                     |
| Force as a third material entity inserted between independently complete objects.                                              | Force is a participant-side readout of the complete joint configuration.            |
| An analyzer-relative component or measured direction already owned by a lone Level-1 object.                                   | Relative components belong to the particle-apparatus relation.                      |
| Relation, capacity, response, accessibility, projector, map, field, or metric automatically treated as a hidden substance.     | Physical interpretation must be independently earned.                               |
| A universal physical $\frac{3}{4}$ , $\frac{1}{4}$ , or other settlement fraction inferred merely from a recurring rank ratio. | A repeated number is not automatically a repeated mechanism.                        |
| Expansion as the unfilled lane of a write or the growth of a finite store of empty addresses.                                  | Any surviving expansion result requires current causal/geometry rederivation.       |
| Separate positive counting of equivalent self-field, source, and response descriptions.                                        | One physical source/sector is counted once.                                         |
| Automatic import of QFT/SM or GR ontology because the same mathematics is used.                                                | Native-first interpretation; correspondence second.                                 |
| V13.13 as a veto against future discovery.                                                                                     | Contradiction opens an audit and may supersede the constitution.                    |

## 12. Known unknowns

An ontology becomes less trustworthy, not more, when it hides the joints where dynamics remain unsolved. V13.13 therefore treats its open problems as part of the constitution.

| ID   | Open joint                            | Question                                                                                                                               | Present status             |
|------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| J-01 | Objective event actualization         | What physically selects or realizes one microscopic event or basin?                                                                    | OPEN                       |
| J-02 | Composition and statistics            | What native local relativistic composition law selects the electron exchange sign and self-composition nullity?                        | OPEN BY ONE DISCRETE LAW   |
| J-03 | Formation and root selection          | What supplies first capture, width/support normalization, winding or branch selection, and formation abundance?                        | OPEN                       |
| J-04 | Native null transfer microdynamics    | What is the native phase/transport law for null relational alternatives without assigning a photon a proper-time clock?                | PARTIAL / OPEN             |
| J-05 | General relational conservation       | Does the exact binary accessibility redistribution generalize to mixed, branch-mixing, and multipartite systems?                       | OPEN                       |
| J-06 | Exteriors and coarse-graining         | How do finite maintained roots join whole-space electromagnetic/gravitational exteriors and many-body macroscopic geometry?            | OPEN                       |
| J-07 | Expansion after the ledger correction | Which V13.2 results survive a rewrite without a literal writable-address substrate?                                                    | RETYPE / REAUDIT REQUIRED  |
| J-08 | Absolute scales and couplings         | What selects support scale, alpha, strong normalization, electroweak scale/orientation, and other dimensional or stiffness interfaces? | OPEN / EXPLICIT INTERFACES |

These are not all the unsolved questions in SFT. They are the open joints most likely to change the ontology itself. A successful actualization theory might add a new physical selector or show that none is needed. A composition theorem might promote structural identity into a full instance algebra. A formation theory might revise the Level-0-to-Level-1 transition. A gravity or expansion rederivation might sharpen the status of geometry. The ontology remains open precisely where the physics remains open.

## 13. The non-ceiling clause

### 13.1 Ground state, not ceiling

V13.13 records the smallest coherent ontology currently justified. It is not an axiom set imposed on future discovery. The fact that a later proposal conflicts with this paper is not evidence that the later proposal is wrong. It is evidence that an ontology challenge must be run.

The priority order is:

physical evidence and successful derivation;

the current ontology register;

historical interpretation and metaphor.

**Frozen means protected from casual drift, not protected from discovery.**

### 13.2 Three ways future work can meet V13.13

**Compatible extension.** A later result adds an object, relation, level, or law not addressed here. The ontology ledger expands, and the new claim is not retroactively attributed to V13.13.



**Refinement or retyping.** A later result shows that a current statement is too broad, too narrow, or incorrectly categorized. The claim is amended, the earlier wording is preserved, and a provenance receipt records the change.

**Contradiction or ontology challenge.** A later result directly conflicts with a current claim. The new result is not rejected for nonconformity. Old and new branches are kept separate, the exact conflict is registered, and hostile review tests whether the disagreement is semantic, scoped, mathematical, or physical.

### 13.3 Ontology challenge protocol

An ontology challenge should:

- register the exact conflicting statements;
- identify the source, status, and scope of the current claim;
- keep old and new branches separate before comparison;
- test whether the conflict is semantic, scoped, mathematical, or physical;
- run hostile regression against load-bearing observations and dependencies;
- preserve the failure if the new proposal fails;
- amend or supersede V13.13 if the new proposal survives;
- retain the earlier claim in the archive as provenance.

A future paper may state:

**ONTOLOGY CHALLENGE:** This program treats SFT V13.13 Section X as a current hypothesis under test. Conflict with that section is an audit trigger, not grounds for automatic rejection.

## 14. Claim boundary and falsification

### 14.1 What V13.13 claims

V13.13 claims that the current SFT stack is coherently interpreted through real causal propagation, persistent recurrent structures, physical joint relations, and irreversible realized history; that no physical cosmic ledger or ontic write operator is part of the current ontology; that the maintained electron is the recurrent structure at Electron V1 scope; that the Level 0/1/2/3 ladder correctly separates causal, intrinsic, relational, and collective facts; that clocks and rulers are downstream of persistent closure; that force is a participant readout of complete joint interaction; that useful mathematics must not be reified without independent physical justification; and that every one of these claims remains amendable by later successful discovery.

### 14.2 What V13.13 does not claim

V13.13 does not claim:

- a new derivation of particle formation, the electron, gravity, expansion, Bell correlations, or time;
- that V13.10 has proven the unique ontology of time;
- that every particle species shares the detailed Electron V1 constitutive architecture;
- that structural identity already derives indistinguishability, exchange antisymmetry, or statistics;
- that the post-interaction persistence criterion is universal;
- that numbers literally do not exist at Level 1;

that the binary relational redistribution theorem is a universal multipartite conservation law;  
 that fields, geometry, metrics, or curvature are unreal;  
 that all V13.1 gravity results have already been rederived in relational language;  
 that V13.2's original writable-address mechanism survives;  
 an objective event selector, a complete transition theory, absolute scale selection, alpha, all couplings,  
 whole-space exteriors, or macroscopic coarse-graining;  
 permission to reject future discoveries that conflict with this document.

### 14.3 Falsification and supersession

A constitutional claim is challenged when a later lawful model or observation requires a physical fact that the claim excludes, or removes a physical fact that the claim treats as necessary. Examples include an independently observable microscopic write substrate, an analyzer-relative component demonstrably possessed by an isolated Level-1 root, a required third force substance not reducible to a complete joint configuration, a particle whose invariant root truly scales continuously with amplitude, or a successful formation law that necessarily uses the completed clock upstream.

The paper is not protected by reinterpretation after the fact. If the conflict survives hostile review, the relevant claim is amended or superseded. The falsification surface is claim-level rather than all-or-nothing: one failed sentence need not erase every independent result in the stack.

## 15. Provenance constitution

### 15.1 Claims, not whole papers

V13.13 adopts no source wholesale. Original V13.8 contains useful projector mathematics and failed ontology. V13.5 contains historical Bell ancestry and superseded write-boundary language. V13.2 contains a valuable global-readout target and a retired writable-address mechanism. V13.12 contains a strong maintained-electron freeze and explicit open exteriors, scale, formation, interactions, and statistics. This is not inconsistency. It is why claim-level provenance is required. [S03, S06, S09, S13]

### 15.2 Every claim needs four labels

Every constitutional or candidate statement should carry:

**source** - which paper, gate, theorem, or assembly decision owns it;

**status** - controlling, model freeze, exact conditional, scoped, candidate, open, retired/retyped, or methodological;

**scope** - the physical domain in which the claim is valid;

**boundary** - what the claim does not establish.

This four-label discipline applies to metaphysics with the same force used for numerical physics. A beautiful ontology sentence is not exempt from provenance merely because it is qualitative.

### 15.3 Supersession must be explicit

A later source may supersede the ontology of an earlier source while retaining its mathematics or negative result. The earlier document remains in the archive. It is not silently rewritten. The later source must identify what changed, what survived, and which source now controls. [S09, S15]

## 15.4 Measurement does not build the ontology

Measurements can identify, calibrate, falsify, or select among structures. They cannot be smuggled into the intrinsic construction and then advertised as predictions. One downstream dimensional anchor is an anchor, not a predicted absolute scale. [S16]

## 16. Source disposition summary

The following summary gives the public inheritance rule. The complete source table appears in Appendix B.

**V13.0-V2:** carry forward real support and open/recurrent branch ancestry; retype literal write machinery.

**V13.1-V3:** carry forward gravity as non-substance and source/geometry response at conditional scope; do not promote unconditional completion or material-substrate imagery.

**V13.2:** preserve the global-readout target; rederive every literal writable-address and capacity-ledger mechanism.

**V13.3 and V13.6:** no separate controlling ontology source identified; no doctrine invented.

**V13.4-V2:** controlling joint-interaction ontology.

**V13.5:** historical entanglement ancestry; superseded for ontology by V13.9-V3.

**V13.7:** action/admissibility ancestry absorbed downstream.

**V13.8-V2:** controlling removal of the ontic write operator and ledger.

**V13.9-V3:** controlling relational model and bounded redistribution theorem.

**V13.10:** scoped definitional proposal for time.

**V13.11:** controlling level/ruler ontology with a firewall around its separable mechanism proposal.

**V13.12:** controlling maintained-electron physics and ontology at Electron V1 scope.

**V13 audit and measurement firewalls:** controlling methodology.

## 17. Authorship, contribution, and source-credit rule

**Shaun Fosmark** - originator, research lead, ontology adjudication, and final claim authority.

**GPT-5.6 SOL** - provenance audit, derivation integration, claim-ledger construction, first-draft assembly, and hostile consistency review.

**Fable** - independent critique, ontology stress testing, structural recommendations, and candidate-extension review.

All three names appear together at the top because V13.13 is an integration produced by the three-way ontology audit and assembly. This does not retroactively reassign authorship of the underlying sector papers. Earlier collaborators remain credited in their source documents, and V13.13 inherits results without erasing their provenance.

## 18. Conclusion

SFT V13.13 is smaller than the metaphors that preceded it.

There is no cosmic ledger waiting to receive the next state. There is no microscopic write operator that makes reality happen. There is no universal present against which every clock is measured. A particle

need not contain a smaller carrier that performs its properties. A force need not be a third substance inserted between two independently complete objects. A metric need not be a rubber medium merely because geometry is physically consequential.

What remains is a compact physical order. Real support propagates causally. Persistent roots recur. Recurrence creates the conditions for identity, rest, tally, and local scale. Relations make separation, components, interaction, measurement, and entanglement meaningful. Many persistent systems make extension and macroscopic apparatus possible. Complete configurations satisfy lawful dynamics. Mathematics represents their restrictions, responses, and readouts.

That order answers why the theory became smaller and larger at the same time. The smallest maintained structure is not isolated from the deepest questions. To say what one electron is, SFT must say where its clock comes from, what its ruler measures, which facts belong to the electron, which belong to its relations, what an interaction changes, what geometry means, and which mathematical objects are descriptions rather than substances.

The ontology is now coherent enough to guide future work and explicit enough to be attacked by it. That is the point of a constitution: not to stop discovery, but to make every future discovery state clearly what it preserves, what it changes, and why.

**SFT is constrained by reality, not by SFT V13.13.**

## References and source key

- [S01] SFT V13.0-V2, *Particles: Light, Mass, Closure, and the Particle-Scale Law - Unified Canonical Edition* (integrated ancestry draft).
- [S02] SFT V13.1-V3, *Gravity Derivation Canon - Receipt-Backed Source of Truth v1.0*.
- [S03] SFT V13.2, *Expansion as Global Time Production and Boundary-Capacity Growth from Irreversible Writes*.
- [S04] V13.3 coverage note: no separately controlling ontology source identified in the audited library under this label.
- [S05] SFT V13.4-V2, *Relational Interactions: Persistent Particle Roots and Complete Joint Configurations - Final v1.0*; controlling derivation Blend v0.11.
- [S06] SFT V13.5, *Entanglement, Bell Correlations, and Settlement-Write Measurement* (historical ontology ancestry).
- [S07] V13.6 coverage note: no separately controlling ontology source identified in the audited library under this label.
- [S08] SFT V13.7, *Master Action II* (action/admissibility ancestry).
- [S09] SFT V13.8-V2, *There Is No Write Operator: From Projector-Compressed Bookkeeping to Lawful Physical Evolution - Final v1.0*.
- [S10] SFT V13.9-V3, *Working Relational Model v1.1 - How Two Things Become Independent - SUPER FREEZE*.
- [S11] SFT V13.10, *What a Clock Is Actually Doing: Time as the Count of Completed History*.
- [S12] SFT V13.11, *The Cones and the Ruler: Where the Scale of the Metric Comes From - v1.0*.
- [S13] SFT V13.12, *C17G19 Parameterized Stable Maintained SFT Electron V1 - Final Physics Freeze*.
- [S14] SFT V13.12 precursor/capstone, *C17G13.1/C17G14 Ontology Capstone - The Particle Is the Structure*.
- [S15] SFT V13 audit doctrine, *Pull-Forward Canon v1.0* and provenance/dependency ledgers.
- [S16] IEMM *Empirical Measurement Firewall v0.1*.

## Appendix A - Claim and provenance ledger

This appendix is the controlling machine-readable ontology ledger rendered inside the public draft. Status and scope remain part of every claim.

| ID   | Short name                              | Statement                                                                                                                                                                                                 | Level      | Status                               | Sources       | Disposition | Boundary                                                                                         |
|------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------|---------------|-------------|--------------------------------------------------------------------------------------------------|
| O-01 | Real physical support                   | Real physical support propagates causally; current SFT models its primitive local propagation as null.                                                                                                    | L0         | CONTROLLING                          | S01; S09; S10 | CORE        | Do not reduce the claim to a purely mathematical trajectory or an imported background substance. |
| O-02 | Open and recurrent branches             | Open null propagation and persistent recurrent massive structure are different physical branches, not different substances.                                                                               | L0→L1      | CONTROLLING                          | S01; S13; S14 | CORE        | The exact formation/selection dynamics remain open.                                              |
| O-03 | Particle is the structure               | A maintained particle is the recurrent physical structure itself, not a carrier inside a container or a photon on a pre-existing track.                                                                   | L1         | CONTROLLING AT ELECTRON V1 SCOPE     | S09; S13; S14 | CORE        | Generalization beyond the maintained-electron class must be earned sector by sector.             |
| O-04 | One-state faces                         | Null tangent/current, support, geometry, source, spin/framing, electromagnetic dressing, recurrence, and clock readout may be different faces of one maintained state rather than independent mechanisms. | L1         | MODEL FREEZE                         | S05; S13; S14 | CORE        | Do not infer that every particle species shares the same detailed constitutive solution.         |
| O-05 | Rest without local stasis               | A persistent rest frame can arise from globally recurrent null propagation whose integrated translational momentum closes.                                                                                | L1         | MODEL FREEZE                         | S13; S14      | CORE        | This is an Electron V1 statement, not a universal particle theorem.                              |
| O-06 | Mass as closure readout                 | Mass is an invariant/readout or response of maintained closure, not a stored substance attached to a carrier.                                                                                             | L1         | CONTROLLING INTERPRETATION           | S01; S13; S14 | CORE        | Absolute mass scale still requires a lawful dimensional anchor or scale theorem.                 |
| O-07 | Closed root is not a scalable amplitude | Branch-preserving added energy normally appears as relative momentum, fields/radiation, or additional channels rather than continuously turning one root into 'more' of the same root.                    | L1/L2      | EARNED AT BRANCH-PRESERVING SCOPE    | S05           | CORE        | Not a complete field-quantization or creation/annihilation theorem.                              |
| O-08 | Structural identity                     | Instances of the same recurrent solution class are expected to share the same intrinsic invariants.                                                                                                       | L1         | CANDIDATE / PLAUSIBLE TARGET THEOREM | S14           | CANDIDATE   | Does not derive indistinguishability, exchange antisymmetry, or statistics.                      |
| O-09 | Amounts versus instances                | Open modes may vary in amplitude/occupancy, whereas persistent closed roots are represented as instances rather than continuously scalable root amplitudes.                                               | L0/L1      | CANDIDATE SYNTHESIS                  | S05; S14      | CANDIDATE   | Requires a native composition theorem before universal promotion.                                |
| O-10 | Independent mass and charge handles     | Mass cadence/closure phase and charge/framing phase are independent physical handles.                                                                                                                     | L1         | EARNED / PRESERVED                   | S09; S05      | CORE        | Numerical charge normalization and charge quantization remain separate questions.                |
| O-11 | No-overwrite retyped                    | Realized four-dimensional history is irreversible: what occurred cannot later become a different event.                                                                                                   | History    | CONTROLLING                          | S09; S10      | CORE        | No occupied-address substrate is implied.                                                        |
| O-12 | No cosmic ledger                        | There is no physical cosmic ledger, memory surface, universal register, or write-address substrate in current SFT ontology.                                                                               | All        | CONTROLLING                          | S09; S10      | CORE        | Ledger and write may survive only as explicitly translated bookkeeping shorthand.                |
| O-13 | No ontic write operator                 | There is no microscopic operator that appends reality; a transition map is the solution map induced by the physical dynamics.                                                                             | All        | CONTROLLING                          | S09           | CORE        | Operator notation remains lawful mathematics.                                                    |
| O-14 | Lawful configuration is primary         | Physical evolution is determined by solving the complete constrained configuration in its lawful domain.                                                                                                  | Dynamics   | CONTROLLING ARCHITECTURE             | S05; S09      | CORE        | The architecture does not itself choose among multiple lawful branches.                          |
| O-15 | Configuration-to-readout hierarchy      | Physical configuration precedes lawful solution; mathematical restriction/response operators and observable readouts come afterward.                                                                      | Method     | CONTROLLING                          | S09           | CORE        | This ordering is a guardrail against reification.                                                |
| O-16 | Null structure has no ruler             | Null causal/conformal structure supplies causal shape but no intrinsic massive rest frame, clock, or metric scale.                                                                                        | L0         | CONTROLLING                          | S12           | CORE        | The claim does not deny mathematically valid vacuum metrics.                                     |
| O-17 | Closure supplies a local unit           | A persistent closed process supplies a rest frame, a recurrence tally, and a local intrinsic unit associated with its Compton scale.                                                                      | L1         | CONTROLLING WITH MECHANISM FIREWALL  | S12; S13      | CORE        | The specific formation mechanism that creates the closure is not thereby solved.                 |
| O-18 | Clock is downstream                     | The first measurable clock is a post-closure recurrence readout and may not be used upstream to select or cause closure.                                                                                  | L1         | CONTROLLING DEPENDENCY RULE          | S13; S14      | CORE        | Protects formation work from circular calibration.                                               |
| O-19 | Time as completed history               | Elapsed time is proposed as the normalized count of completed local recurrent history rather than an external universal flow.                                                                             | L1/History | DEFINITIONAL PROPOSAL                | S11           | SCOPED      | V13.10 explicitly does not claim a derivation of time.                                           |
| O-20 | No master clock or universal present    | There is no master clock or universal present shared across the universe.                                                                                                                                 | L1/L2      | CONTROLLING RELATIONAL CONSEQUENCE   | S10; S11      | CORE        | Cross-clock statements require actual comparison relations.                                      |

| ID   | Short name                                     | Statement                                                                                                                                                                                     | Level | Status                                           | Sources            | Disposition  | Boundary                                                                                                          |
|------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------|--------------------|--------------|-------------------------------------------------------------------------------------------------------------------|
| O-21 | A clock makes its own elapsed time             | A clock does not estimate one external duration; its completed recurrence is its own elapsed-time history.                                                                                    | L1    | DEFINITIONAL PROPOSAL                            | S11                | SCOPED       | Instrument reading and comparison errors remain possible even though the clock's own history is local.            |
| O-22 | Level 0                                        | Level 0 contains null propagation, light cones, causal order, conformal geometry, and helicity, but no intrinsic ruler or massive clock.                                                      | L0    | CONTROLLING                                      | S12                | CORE         | Do not assign Level-1 or Level-2 objects upstream.                                                                |
| O-23 | Level 1                                        | Level 1 begins with a persistent closed process and contains root identity, rest frame, local tally/unit, mass, charge, and spin magnitude.                                                   | L1    | CONTROLLING                                      | S12; S13; S14      | CORE         | Analyzer-relative or separation-dependent quantities are excluded.                                                |
| O-24 | Level 2                                        | Level 2 begins when persistent systems participate together; separation, relative motion/orientation, interaction, entanglement, and measurement become meaningful.                           | L2    | CONTROLLING                                      | S10; S12; S05      | CORE         | A relation is not automatically a new material substance.                                                         |
| O-25 | Level 3                                        | Many persistent systems support rigid extension, rods, collective gravitation, thermodynamics, horizons, and other macroscopic structures.                                                    | L3    | CONTROLLING LADDER STATEMENT                     | S12                | CORE         | Detailed emergence laws remain sector-specific.                                                                   |
| O-26 | Measurement is comparison                      | Measurement exists at Level 2 because measurement is a physical comparison between participants.                                                                                              | L2    | CONTROLLING                                      | S12; S10           | CORE         | This is not observer-created reality or human-dependent actualization.                                            |
| O-27 | Magnitude versus component                     | A single persistent object may possess invariant magnitudes, while a direction/component relative to an analyzer or another object belongs to the relation.                                   | L1/L2 | CONTROLLING                                      | S10; S12           | CORE         | Do not hide relational components inside an independently complete local answer.                                  |
| O-28 | Calibration begins at relation                 | An intrinsic magnitude may exist at Level 1, but a numerical laboratory value and unit assignment require comparison to a physical standard.                                                  | L1/L2 | CANDIDATE SYNTHESIS                              | S12; S16           | CANDIDATE    | Stronger wording such as 'numbers begin at two' is not presently justified.                                       |
| O-29 | Relations can be physical                      | Some facts belong irreducibly to a joint physical configuration rather than to either participant separately.                                                                                 | L2    | CONTROLLING                                      | S10                | CORE         | Relation, capacity, and accessibility are not substances or signals.                                              |
| O-30 | Joint continuation need not factor             | After a genuine interaction, the lawful pair continuation need not equal the product of two independently complete isolated continuations.                                                    | L2    | EARNED / MODEL FREEZE                            | S10; S05           | CORE         | Nonfactorization alone is not sufficient for quantum entanglement.                                                |
| O-31 | Separation does not erase relation             | Clean invertible local transport can preserve nonfactorizable relational structure after direct mechanical overlap ends.                                                                      | L2    | EARNED IN V13.9-V3 DOMAIN                        | S10                | CORE         | Interaction with distinguishing environments can alter accessibility.                                             |
| O-32 | Entanglement as Level-2 structure              | Entanglement is modeled as phase-bearing joint structure rather than a payload independently carried by either participant.                                                                   | L2    | SUPER-FREEZE MODEL CLAIM                         | S10                | CORE         | General mixed-state and multipartite theory remains open.                                                         |
| O-33 | Scoped relational redistribution               | In the normalized binary pure branch-preserving distinguishing domain, direct pair accessibility and enlarged-system relation obey an exact squared-capacity redistribution identity.         | L2/L3 | EXACT CONDITIONAL                                | S10                | SCOPED       | Do not promote to universal relational conservation.                                                              |
| O-34 | Relation is not a transmitted substance        | Relational structure is physically significant without being a fluid, signal, hidden field, conserved charge, or material sent between particles.                                             | L2    | CONTROLLING                                      | S10                | CORE         | Causal propagation of real support is still required.                                                             |
| O-35 | Force is not a third substance                 | An interaction does not require two independently complete objects plus a third entity called force.                                                                                          | L2    | CONTROLLING                                      | S05                | CORE         | Fields or geometry may still be real parts of the complete configuration.                                         |
| O-36 | Force as participant readout                   | Force is the participant-side change in lawful continuation produced by the complete joint configuration relative to isolated continuation.                                                   | L2    | CONTROLLING                                      | S05                | CORE         | Proper force need not capture gravitational free fall.                                                            |
| O-37 | Root-surplus distinction                       | Persistent root invariant content is distinct from surplus relational, kinematic, field, radiation, or channel content.                                                                       | L1/L2 | EARNED AT BRANCH-PRESERVING SCOPE                | S05                | CORE         | Formation and identity-changing transitions remain separate.                                                      |
| O-38 | Interaction as joint solve                     | Real interactions must be solved as joint physical states with isolated roots as an asymptotic boundary, not as an invisible external force applied to permanently isolated objects.          | L2    | CONTROLLING FIREWALL                             | S05; S13           | CORE         | Full finite-overlap solves remain open.                                                                           |
| O-39 | One architecture, different constitutive plugs | A common constrained-joint architecture does not imply identical microscopic physics for electromagnetic, strong, weak, gravity, detector, or other sectors.                                  | L2    | CONTROLLING ARCHITECTURE                         | S05                | CORE         | Sector-specific domains, kernels, selectors, and normalizations remain explicit.                                  |
| O-40 | Gravity source versus mutual gravity           | A persistent root may possess mass/source and local closure-gravity structure, while mutual acceleration, redshift, deflection, and other gravitational observations are relational readouts. | L1/L2 | CONTROLLING DISTINCTION / GRAVITY REAUDIT TARGET | S12; S13; S05; S02 | CORE + AUDIT | Do not yet claim that spacetime curvature is unreal or that all exterior gravity has been rederived relationally. |
| O-41 | Geometry is structure, not material substrate  | Coordinates, frames, rods, clocks, metric scale, and macroscopic extension are constituted from causal structure and persistent systems rather than supplied by a prior material stage.       | L0-L3 | CONTROLLING INTERPRETATION                       | S12; S09           | CORE         | This does not deny geometry, mathematically valid metrics, or physical field support.                             |



| ID   | Short name                             | Statement                                                                                                                                                      | Level      | Status                      | Sources                          | Disposition | Boundary                                                                                                 |
|------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------|----------------------------------|-------------|----------------------------------------------------------------------------------------------------------|
| O-42 | Post-interaction persistence criterion | A candidate new-thing criterion is that a temporary joint configuration captures into a lawful continuation that persists after the initiating encounter ends. | Transition | CANDIDATE GENERAL PRINCIPLE | S05; S13; S14                    | CANDIDATE   | The first-capture law and claimed universality across chemistry, records, and phases are not yet earned. |
| O-43 | Claim-level provenance                 | Claims and sections—not whole documents—are the unit of inheritance; superseded papers may contain portable mathematics and failed ontology at once.           | All        | CONTROLLING METHOD          | S15; S09                         | METHOD      | No document is adopted or rejected wholesale merely by version number.                                   |
| O-44 | Native-first correspondence            | Use SFT-native interpretation first; QFT/SM or GR terminology is a precision/correspondence layer unless SFT independently earns the same ontology.            | All        | CONTROLLING METHOD          | S05                              | METHOD      | Standard mathematical names remain allowed when the structures genuinely coincide.                       |
| O-45 | V13.13 is not a ceiling                | V13.13 records the smallest coherent ontology currently justified; it does not legislate what future reality is allowed to contain.                            | All        | ASSEMBLY RULE               | Assembly decision                | NON-CEILING | Future work may challenge any statement through explicit audit.                                          |
| O-46 | Supersession by discovery              | A later result that survives hostile review may amend or supersede V13.13; the old claim remains archived with explicit provenance.                            | All        | ASSEMBLY RULE               | Assembly decision; S15 precedent | NON-CEILING | Contradiction triggers audit, not automatic rejection of the new result.                                 |

## Appendix B - Source disposition map

A numbered paper is not adopted or rejected wholesale. This map states exactly what each source may contribute to V13.13 and what may not be promoted from it.

| ID  | Version  | Authority                                                                | Disposition                             | Carry forward                                                                                                                                                              | Do not promote                                                                                                                           |
|-----|----------|--------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| S01 | V13.0-V2 | Integrated ancestry / draft; not current ontology controller             | ANCESTRY / RETYPE                       | Real physical support; open-null versus persistent-recurrent branches; conservation-compatible formation; branch rigidity.                                                 | Literal write/ledger language, obsolete write-lane provenance, or a completed microscopic formation law.                                 |
| S02 | V13.1-V3 | Conditional gravity derivation canon                                     | SCOPED CONDITIONAL SOURCE               | Gravity is not an added substance; geometry/source response is tied to real support; field-equation claims remain explicitly conditional.                                  | An unconditional gravity completion, a proof that a lone particle literally bends a material substrate, or obsolete ledger ontology.     |
| S03 | V13.2    | Historical V13.2 mechanism freeze                                        | RETYPE / CURRENT RE-DERIVATION REQUIRED | Expansion is not a new force, medium, or dark substance; preserve the global-readout target and stock/flow/capacity distinctions only where independently re-earned.       | Writable-address budget, cosmic ledger, literal fresh-address mechanism, or expansion as a universal write-lane split.                   |
| S04 | V13.3    | Coverage note                                                            | NO INVENTED DOCTRINE                    | Nothing added merely to make the version sequence look continuous.                                                                                                         | Any reconstructed V13.3 ontology without a controlling source.                                                                           |
| S05 | V13.4-V2 | Controlling public interaction narrative; Blend v0.11 controls physics   | CONTROLLING                             | Force as participant-side readout; complete constrained joint configurations; root-surplus distinction; constitutive-plug diversity; explicit child programs.              | Force as a third substance, a completed Standard Model derivation, or universal sector-specific closure beyond the declared scope.       |
| S06 | V13.5    | Historical quantum ancestry                                              | SUPERSEDED FOR ONTOLOGY                 | Historical Bell and no-signaling ancestry where later V13.9-V3 explicitly preserves it.                                                                                    | Shared write-boundary, outcome-write, hazard, or ledger ontology as controlling.                                                         |
| S07 | V13.6    | Coverage note                                                            | NO INVENTED DOCTRINE                    | Nothing added without a source.                                                                                                                                            | Any reconstructed V13.6 ontology merely from numbering.                                                                                  |
| S08 | V13.7    | Action/admissibility ancestry                                            | ANCESTRY                                | Action-domain and admissibility architecture where absorbed by V13.4-V2 and V13.8-V2.                                                                                      | Its older additive sector decomposition or unsettled branch machinery as current ontology.                                               |
| S09 | V13.8-V2 | Controlling ontology for write/operator/projector sector                 | CONTROLLING                             | No ontic write operator or ledger; no-overwrite as irreversible realized 4D history; operators as mathematical response/restriction; lawful-configuration hierarchy.       | Universal settlement lanes, ontic trace fractions, or obsolete write-lane explanations.                                                  |
| S10 | V13.9-V3 | Controlling research model for V13.9 replacement                         | CONTROLLING                             | Level-2 relational ontology; joint nonfactorization; clean-separation persistence; scoped accessibility redistribution; causal locality without Bell-local separability.   | Universal multipartite conservation, complete composition, objective event realization, or transition dynamics beyond the frozen domain. |
| S11 | V13.10   | Definitional proposal with stated consequences                           | PROPOSAL / SCOPED                       | No universal present or master clock; local completed-history account of elapsed time; clock-as-maker framing.                                                             | The proposal as an established theorem, or its older literal queue/ledger wording without V13.8-V2 translation.                          |
| S12 | V13.11   | Controlling level/ruler ontology; mechanism section explicitly separable | CONTROLLING WITH SECTION FIREWALL       | Level 0/1/2/3 ladder; null structure lacks a ruler; persistent closure supplies a local unit; measurement is comparison; apparatus is constituted rather than presupposed. | The proposed null-carrier closure mechanism in its separable section as an independently established theorem.                            |

| ID  | Version                   | Authority                                             | Disposition                      | Carry forward                                                                                                                                    | Do not promote                                                                                                      |
|-----|---------------------------|-------------------------------------------------------|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| S13 | V13.12                    | Controlling maintained-electron physics freeze        | CONTROLLING AT ELECTRON V1 SCOPE | One finite recurrent maintained electron state; post-closure clock; one-count source accounting; explicit open interfaces and no-retuning rules. | Absolute scale, intrinsic alpha, formation, full exteriors, interactions, or exchange/statistics as solved.         |
| S14 | V13.12 precursor/capstone | Controlling ontology capstone within electron program | CONTROLLING + CANDIDATE TARGETS  | Particle equals recurrent structure; no photon inside a container; cadence is a readout; structural-identity route.                              | Structural identity as a full indistinguishability or spin-statistics theorem.                                      |
| S15 | V13 audit doctrine        | Methodological audit guardrail                        | CONTROLLING METHOD               | Claims and sections, not whole documents, are the unit of inheritance; provenance is explicit; disagreement becomes a named audit question.      | Whole-document adoption, normalization by resemblance, or archive material as an automatic solver of current opens. |
| S16 | Measurement firewall      | Frozen reconstruction methodology                     | CONTROLLING METHOD               | Measurements may identify or test a structure, not build it; one later dimensional anchor is an anchor, not a prediction.                        | Empirical target leakage into intrinsic ontology or selection.                                                      |

## Appendix C - Candidate extension register

These claims are intentionally preserved in the paper without being promoted to the constitutional core. Each has an explicit promotion and kill/retype gate.

| ID   | Candidate                              | Proposed sentence                                                                                                                                                                                                               | Status                            | Promotion gate                                                                                                                                              | Kill / retype gate                                                                                                                                       | Placement                                                                                                 |
|------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| C-01 | Post-Interaction Persistence Criterion | A new persistent thing is formed when a temporary joint configuration captures into a lawful continuation that remains self-maintaining after the initiating encounter ends.                                                    | CANDIDATE GENERAL PRINCIPLE       | Derive a domain-independent criterion and verify it without changing definitions across particle formation plus at least one nonparticle persistence class. | Different sectors require incompatible notions of persistence, or the criterion circularly uses a downstream clock/identity to create itself.            | Candidate transition principle; not in the constitutional core until promoted.                            |
| C-02 | Structural Identity and Instancehood   | Identical roots are one recurrent solution class instantiated many times; same-class instances share invariant root properties.                                                                                                 | PLAUSIBLE TARGET THEOREM          | Prove uniqueness/equivalence of the relevant invariant root class and show instance-independent intrinsic readouts.                                         | Multiple inequivalent hidden branches reproduce the same public root label with different intrinsic invariants.                                          | Identity chapter with an explicit firewall against claiming exchange statistics.                          |
| C-03 | Open Amounts / Closed Instances        | Open modes come in continuously variable amounts; persistent closed roots come in discrete instances, while surplus energy occupies motion, fields, radiation, or new channels.                                                 | SCOPED SYNTHESIS                  | Native creation/composition law reproduces the claimed open/closed asymmetry without importing the result from QFT ontology.                                | A lawful closed-root family contains a physical continuous root-amplitude degree of freedom that is not merely relational or excitation content.         | Identity/instancehood chapter as a scoped candidate.                                                      |
| C-04 | Magnitude at One, Calibration at Two   | One persistent object may possess an intrinsic magnitude; a numerical measured value requires a comparison relation and a physical standard.                                                                                    | CANDIDATE ONTOLOGY SYNTHESIS      | Formalize intrinsic dimensionless invariants, scale transport, and measurement maps without erasing one-object facts.                                       | A genuine operational number is shown to be fully intrinsic without any relation to a unit, standard, or other physical structure.                       | Measurement/scale section; replaces the overstrong phrase 'numbers begin at two.'                         |
| C-05 | Relational Reading of Curvature        | A root carries source and intrinsic response structure; what is observed as bending, deflection, redshift, or mutual gravitation belongs to the lawful relational configuration rather than to a deformable material substrate. | HIGH-VALUE GRAVITY REAUDIT TARGET | Reformulate the V13.1 exterior and two-body observables in the current joint-configuration ontology while preserving all earned gravity limits.             | A required independently measurable geometric degree of freedom exists even when no relational/source interpretation can reproduce its physical content. | Geometry/gravity section as an explicit research target, not a declaration that curvature does not exist. |

## Appendix D - Open joints register

These are the unresolved joints most likely to change or extend the ontology. Their visibility is part of the claim boundary, not a weakness to be hidden.

| ID   | Open joint                    | Question                                                                                                        | Owner                                               | Status                   | Ontology risk                                                                             |
|------|-------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------|
| J-01 | Objective event actualization | What physically selects or realizes one microscopic event/basin?                                                | Native Transition Dynamics / detector actualization | OPEN                     | Do not smuggle a collapse substance, hidden selector, or ontic write event into the core. |
| J-02 | Composition and statistics    | What native local relativistic composition law selects the electron exchange sign and self-composition nullity? | P6-L root-composition / spin-statistics program     | OPEN BY ONE DISCRETE LAW | Structural identity is not yet quantum indistinguishability or Fermi statistics.          |
| J-03 | Formation and root selection  | What supplies first capture, width/support normalization, winding/branch selection, and formation abundance?    | Particle formation / Native Transition Dynamics     | OPEN                     | Do not use the post-closure clock or frozen maintained state upstream to cause formation. |

| ID   | Open joint                            | Question                                                                                                                                    | Owner                                              | Status                     | Ontology risk                                                                                           |
|------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------|
| J-04 | Native null transfer microdynamics    | What is the native phase/transport law for null relational alternatives without assigning a photon a proper-time clock?                     | V13.9 transition/phase child program               | PARTIAL / OPEN             | Avoid photon rest-frame language and hidden carried clocks.                                             |
| J-05 | General relational conservation       | Does the exact binary accessibility redistribution generalize to mixed, branch-mixing, and multipartite systems?                            | V13.9 general composition / CKW program            | OPEN                       | Do not universalize a theorem beyond its binary pure branch-preserving domain.                          |
| J-06 | Exteriors and coarse-graining         | How do finite maintained roots join whole-space electromagnetic/gravitational exteriors and many-body macroscopic geometry?                 | Exterior EM/gravity and macro-gravity programs     | OPEN                       | Do not retune the internal particle merely to hit a far-field observable.                               |
| J-07 | Expansion after the ledger correction | Which V13.2 expansion results survive when rewritten through current causal/geometry dynamics without a literal writable-address substrate? | Expansion/cosmology rederivation                   | RETYPE / REAUDIT REQUIRED  | No cosmic ledger, universal write capacity, or unfilled continuation lane may be reintroduced by habit. |
| J-08 | Absolute scales and couplings         | What selects support scale, alpha, strong normalization, electroweak scale/orientation, and other dimensional or stiffness interfaces?      | Formation, coupling, and relational-scale programs | OPEN / EXPLICIT INTERFACES | One successful local structure must not be inflated into an absolute all-sector derivation.             |

SFT is constrained by reality, not by SFT V13.13.